DAKINI AI CRM

Enterprise Platform Documentation & Comprehensive System Manual

Detailed module specifications, Layer 2 Edge API Gateway, Supabase PostgreSQL data layer, and end-to-end user workflows.

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Architecture Stack:	React 18 SPA | Python FastAPI Layer 2 Gateway | Supabase PostgreSQL
Security Standard:	HMAC-SHA256 JWT, E1 Edge Shield, RBAC Multi-Tenancy
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1. Executive Architecture & 3-Tier System Design

Dakini CRM implements an enterprise 3-tier architecture separating presentation, edge security/routing, and domain data persistence. This ensures horizontal scalability, zero direct client-to-database exposure, and strict row-level security.

Architecture Layer	Technology Stack	Core Responsibility
Layer 1: Frontend SPA Port 3000	React 18, Vite, React Router v6, Lucide Icons	Interactive client SPA, URL-based stateful navigation, Kanban board, dynamic TopBar sync, Day/Week/Month calendar.
Layer 2: Edge Gateway Port 8080	Python FastAPI, PyJWT, Starlette, Uvicorn	Reverse proxy (/v1/*), JWT token issuance/verification, sliding-window rate limiting, request size caps, correlation tracing.
Layer 3: Application & Data Port 8000 & Supabase	FastAPI, SQLAlchemy 2.0 Async, Psycopg3, Supabase DB	Business domain logic, multi-tenant row-level isolation (tenant_id), PostgreSQL pooler connection, schema validations.

2. Authentication, Roles & Tenant Isolation (RBAC)

Dakini CRM implements strict cryptographic authentication. When a user authenticates via POST /v1/auth/token, the Gateway issues a signed HMAC-SHA256 JWT containing user ID, tenant ID, and role assignments.

Role Tier	Default User	Email Credentials	Permissions & Scopes
Owner	Elena Rostova	elena.rostova@dakini.ai	Full organization administrator, settings customization, user quota assignments, all pipelines and reports.
Manager	Marcus Vance	marcus.vance@dakini.ai	Sales team supervisor, pipeline approvals, assigning reps to incoming leads, sales targets tracking.
Sales Rep	Dev Singhania / Asha Nair	dev.singhania@dakini.ai	Field sales representative, lead qualification, deal progression, task logging, client communication.

3. Core Sales Modules & Workflows

Dashboard (/dashboard)

Visual command center featuring real-time KPI metric tiles (ARR, Open Pipeline, Win Rate, Stage Velocity), interactive revenue charts, and live Supabase synchronization status.

Leads Management (/leads & /leads/:id)

Multi-column lead table with AI Lead Scoring (0-150), lead stage filters (New, Contacted, Qualified, Disqualified), CSV import/export, and 1-click 'Convert Lead' modal creating an Account, Contact, and Deal simultaneously.

Deals & Pipeline (/deals)

Visual Kanban board displaying deals across 5 standardized stages: Qualified -> Proposal -> Negotiation -> Won -> Lost. Supports drag-and-drop progression with automated quota attainment updates.

Accounts & Contacts (/accounts, /contacts)

Enterprise account directories storing company domains, industry verticals, ARR health indicators, and associated executive contacts.

Campaigns (/campaigns)

Omnichannel outreach tracker managing campaign budgets, actual spend, pipeline generated, and calculated ROI percentages across Roadshows, Webinars, and Outbound sequences.

Tasks & To-Dos (/tasks)

Activity tracker linking tasks directly to customer records with priority flags (High, Medium, Low), due dates, and 1-click completion checkboxes.

4. Engagement & Communication Hub (/inbox)

The Communication Hub implements an enterprise 3-pane layout inspired by modern mail clients, directly connecting customer emails to matching CRM records.

Folder	Functionality & Features
Inbox	Incoming customer email threads linked to Lead and Contact records. Includes 1-click 'Dakini Smart Reply' generating AI responses based on CRM context.
Sent	Dispatched agreements and correspondence with read receipts and delivery confirmations ('Delivered & Read - Opened 3 times').
Drafts	Autosaved drafts with full inline editor, discard options, and 'Dakini AI Enhance' tone polishing.
Templates	Enterprise cadence templates (Inbound Demo Follow-Up, Enterprise MSA Package, Pipeline Reviver, Executive Sponsor Briefing). Includes interactive Merge Variable Playground ({{first_name}}, {{company}}), Copy Template to Clipboard, and 'Use In New Email' draft creator.

5. Calendar & Engagement Scheduling (/calendar)

The Calendar module offers 3 dynamic views designed for sales rep meeting management and executive cadence reviews:

• **Day View:** Detailed hourly schedule (8:00 AM – 6:00 PM) displaying customer demos, meeting types, attendee pills, direct 'Join Video Bridge' buttons, and open slot indicators.

• **Week View:** 7-column Monday through Sunday schedule displaying event cards, company affiliations, and highlighted active today indicators.

• **Month View:** 35-day October 2026 calendar grid with date numbers (1 to 31), meeting pills, and 1-click jump-to-day actions.

6. Autonomous AI Copilot & Workflow Engine

Dakini CRM integrates AI directly into rep workflows rather than as an isolated chatbot:

• **Dakini AI Chat Dock:** Slide-out dock answering pipeline queries ('Which deals are closing this month?'), drafting custom emails, and recommending next steps.

• **Autonomous Lead Scoring:** Telemetry algorithms automatically score incoming prospects (e.g. Aarav Patel scored at 128 based on demo requests and API sandbox usage).

• **Executive Approval Queue:** Safeguard preventing autonomous agents from dispatching contract discounts without manager signoff.

• **Workflow Automations (/automations):** Visual triggers (e.g. 'Lead Score > 90 -> Auto-assign to Enterprise Rep & Notify Slack War Room').

7. Layer 2 Edge / API Gateway Modules (E1–E8 Specifications)

The Python FastAPI Gateway implements all 8 Low-Level Design (LLD) edge security modules running on port 8080:

Module	Implementation File	Function & Security Enforcement
E1: Edge Shield	`gateway/middleware/edge_shield.py`	Origin secret validation (X-Edge-Secret), cache-busting headers on dynamic APIs.
E2: Health Probes	`gateway/routers/health.py`	Non-blocking /health and /v1/health probing Layer 3 and Supabase readiness.
E3: Versioned Proxy	`gateway/routers/proxy.py`	Reverse-proxies /v1/* to Layer 3, forbids mock responses in production.
E4: Token Auth	`gateway/security/auth.py`	Issues and verifies HMAC-SHA256 JWTs, validates exp, forwards user/tenant claims.
E5: Rate Limiter	`gateway/middleware/rate_limiter.py`	Sliding-window limiter returning 429 Too Many Requests and Retry-After.
E6: Request Validation	`gateway/middleware/request_validation.py`	Enforces 16KB header cap, 1MB payload cap, method whitelist, application/json Content-Type.
E7: Correlation Logging	`gateway/middleware/correlation.py`	Injects X-Correlation-ID (req_) for end-to-end distributed tracing.
E8: Error Normalization	`gateway/errors.py`	Standardized JSON error envelopes across all HTTP codes (BAD_GATEWAY, UNAUTHORIZED).

8. Database Administration & Supabase Verification

All customer records are permanently persisted in **Supabase PostgreSQL** cloud. Direct database connectivity is managed with connection pooling and statement timeouts.

Live Proof & Verification Commands:

To verify that the frontend, gateway, and database are in 100% synchronization, developers and architects can run automated test scripts:

Command	Expected Output / Verification Result
`pytest -q`	Runs all 59 automated tests (27 Layer 2 + 32 Layer 3). Confirms 100% test pass rate.
`python scripts/live_proof.py`	Compares raw SQL direct from Supabase vs API Gateway JSON. Proves 100% data match.

`python scripts/verify_live_api.py`	Executes live CRUD cycles creating, reading, and soft-deleting leads and deals.

Engineering Sign-off: *Verified & Approved for Enterprise Deployment.*
All Layer 1 User Channels, Layer 2 Edge Gateway, and Layer 3 Supabase Data integrations are operational.